MOHAMED EISSA ABDELKAREEM

Computer Science Graduate · IoT & Cloud Solutions

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PROFILE

Computer Science graduate (2024) building connected IoT devices and cloud-hosted applications. Built Wanas, an end-to-end ESP32-S3 device that captures voice and touch input and calls a cloud LLM API over Wi-Fi (HTTPS/REST). Delivered a client booking and payments platform to production on managed cloud services, and gained network support and IT operations experience through internships and military service. Works directly with clients and coordinated cross-functional teams as Head of PR at GDSC; eager to grow in IoT connectivity, M2M and cloud solutions.

EDUCATION & CERTIFICATIONS

B.Sc. Computer Science — Future University in Egypt (FUE), affiliated with University of Cincinnati, USA | Oct 2020 – Aug 2024

- **Graduation Project:** MAVS — Multi-Modal Authentication and Verification System (Arabic OCR and data-extraction module).
- **Certification:** Cisco Networking Academy — CCNAv7: Introduction to Networks (Mar 2024).
- **In progress:** AWS Skill Builder — AWS Cloud Practitioner Essentials.

TECHNICAL SKILLS

- **IoT & Embedded:** ESP32-S3, Arduino, sensors, GPIO, SPI, I2S, Wi-Fi, C/C++ (fundamentals), prototyping, power budgeting.
- **IoT Connectivity (fundamentals):** M2M, MQTT, HTTP/REST, cellular IoT (2G–5G, NB-IoT, LTE-M), LPWAN (LoRaWAN), BLE.
- **Networking:** TCP/IP, IPv4/IPv6 addressing, subnetting, routing and switching basics, LAN troubleshooting, Packet Tracer.
- **Cloud:** Supabase (managed PostgreSQL), Vercel, IaaS/PaaS/SaaS models, AWS core services (fundamentals).
- **Data & Programming:** Python, SQL (MySQL, PostgreSQL), Java, OOP, Power BI (basic).
- **Delivery & Tools:** requirements gathering, technical documentation, Agile/Scrum and Waterfall (fundamentals), Git, Linux.

KEY PROJECTS

Wanas — Connected AI Desk Device (IoT, ESP32-S3) | Independent Project, 2026

- End-to-end flow: voice/touch input → ESP32-S3 → Wi-Fi → cloud LLM API (Gemini, HTTPS/REST) → audio and display output.
- Integrated a GC9A01 SPI display, INMP441 MEMS mic, MAX98357A Class-D amp (two I2S ports) and TTP223 touch sensor.
- Calculated the 3.3V/5V power budget, added capacitors to prevent Wi-Fi brownout resets, and documented the wiring map.
- *Stack:* C/C++, ESP32-S3, Wi-Fi, HTTPS/REST, SPI, I2S, Gemini API

Shadia Travel — Tourism Booking Platform (Mobile & Web) | Freelance, 2026

- Gathered client requirements and delivered a mobile app, admin dashboard and web portal (one monorepo) to production.
- Built booking and payment flows (Paymob) on a managed cloud backend (Supabase/PostgreSQL), deployed on Vercel.
- *Stack:* Flutter (Dart), Next.js (TypeScript), Supabase, PostgreSQL (RLS), Paymob, Vercel

Other Projects

- **Smart Parking & Smart Gate System** (University): Arduino system using sensors to detect vehicles and control the gate.
- **MAVS — Arabic OCR Module** (Graduation Project): extracted and validated ID card fields into MySQL (OpenCV, Tesseract).

EXPERIENCE

Freelance Developer — Cairo | Mar 2026 – Present

- Work directly with clients to scope requirements, estimate effort and cost, and deliver projects end to end.

IT Support Intern — Bayan Tech, Cairo | Apr 2026 – Jun 2026

- Provided hardware, software and end-user support across the office.

IT Help Desk (Military Service) — Military Engineers Dept., Systems Branch, Cairo | Mar 2025 – Mar 2026

- Improved a paper-based process with a digital archive and tracking system for administrative documents and estimations.
- Maintained branch network infrastructure and resolved connectivity, hardware and software issues for users.

Cyber Security Trainee — Ministry of Military Production (Training Sector), Cairo | Jul 2023 – Aug 2023

- Hands-on vulnerability assessment with Burp Suite and Metasploit; practiced applied cryptography fundamentals.

Network Support Trainee — Misr Phosphate Company, Cairo | Aug 2021 – Sep 2021

- Supported network connectivity and troubleshooting; simulated network topologies in Cisco Packet Tracer.

LEADERSHIP, ACTIVITIES & LANGUAGES

- Head of PR — GDSC FUE and League of Arab States Integrated Model; member of IEEE at FUE.
- Delivered C++ peer-learning sessions and workshops for junior students at FUE.
- **Languages:** Arabic (native) · English (fluent).